Mixed Numbers in $\mathbb{Q}$

1. Convert $3\frac{2}{5}$ into an improper fraction.
2. Convert $\frac{19}{4}$ into a mixed number.
3. Convert $-4\frac{3}{7}$ into an improper fraction.
4. Convert $-\frac{29}{6}$ into a mixed number.
5. Convert $6\frac{5}{8}$ into an improper fraction.
6. Convert $-\frac{53}{12}$ into a mixed number.
7. A recipe calls for $\frac{11}{4}$ cups of milk. Express this quantity as a mixed number of cups.
8. A carpenter measures a beam of length $5\frac{3}{8}$ feet. To cut equal-sized shims, the carpenter needs to express this length as an improper fraction in eighths of a foot. What is the length as an improper fraction?
9. A delivery vehicle's cargo weight exceeded its limit by $-\frac{37}{5}$ kilograms relative to target capacity. Write this fraction as a mixed number.
10. A marathon runner completed a training session in $2\frac{7}{10}$ hours. How many tenths of an hour does this total duration represent as an improper fraction?

Solutions

1. $\frac{17}{5}$

   Explanation: Multiply the integer part by the denominator and add the numerator:

   $$3\frac{2}{5} = \frac{(3 \times 5) + 2}{5} = \frac{15 + 2}{5} = \frac{17}{5}.$$

2. $4\frac{3}{4}$

   Explanation: Divide 19 by 4 to find the integer quotient and remainder ($19 = 4 \times 4 + 3$):

   $$\frac{19}{4} = 4\frac{3}{4}.$$

3. $-\frac{31}{7}$

   Explanation: Keep the negative sign outside while converting the magnitude:

   $$-4\frac{3}{7} = -\left(\frac{(4 \times 7) + 3}{7}\right) = -\left(\frac{28 + 3}{7}\right) = -\frac{31}{7}.$$

4. $-4\frac{5}{6}$

   Explanation: Divide 29 by 6 to get quotient 4 with remainder 5, preserving the negative sign ($29 = 4 \times 6 + 5$):

   $$-\frac{29}{6} = -4\frac{5}{6}.$$

5. $\frac{53}{8}$

   Explanation: Multiply the integer part by the denominator and add the numerator:

   $$6\frac{5}{8} = \frac{(6 \times 8) + 5}{8} = \frac{48 + 5}{8} = \frac{53}{8}.$$

6. $-4\frac{5}{12}$

   Explanation: Divide 53 by 12 to get quotient 4 with remainder 5 ($53 = 4 \times 12 + 5$):

   $$-\frac{53}{12} = -4\frac{5}{12}.$$

7. $2\frac{3}{4}$ **cups**

   Explanation: Convert the improper fraction to a mixed number by dividing 11 by 4 ($11 = 2 \times 4 + 3$):

   $$\frac{11}{4} = 2\frac{3}{4} \text{ cups}.$$

8. $\frac{43}{8}$ **feet**

   Explanation: Convert the mixed number to an improper fraction:

$$5\frac{3}{8} = \frac{(5 \times 8) + 3}{8} = \frac{40 + 3}{8} = \frac{43}{8} \text{ feet.}$$

9. $-7\frac{2}{5}$ **kg**

   Explanation: Divide 37 by 5 to get quotient 7 with remainder 2, retaining the negative sign:

$$-\frac{37}{5} = -7\frac{2}{5} \text{ kg.}$$

10. $\frac{27}{10}$ **hours**

    Explanation: Convert the mixed number duration to an improper fraction:

$$2\frac{7}{10} = \frac{(2 \times 10) + 7}{10} = \frac{20 + 7}{10} = \frac{27}{10} \text{ hours.}$$